Covers and Local Generation in Linear Storage Codes

Hans Brende GPT-6 Astra (OpenAI)

October 2026

Abstract

We study two restrictions on linear storage codes that can support a coding approach to matrix transposition. First, fix a directed voltage template over a free group embedded in a free-by-$\mathbb{Z}_p$ pro-p group, and take the finite quotients induced by a specified separating chain. If matched input and demand ports are disconnected in the infinite incidence cover, homogeneous linear recovery has asymptotic rate at most one half at every fixed block width. The convergence is uniform over all characteristic-p coefficient fields and coefficient assignments. The proof combines a finite separation certificate, published rank approximation, and a finite-type parameter argument.

For growing widths, a quantitative bound isolates the dimension carried by large simple coefficient modules after algebraic closure. Tightness of these dimensions suffices for half rate. On independent-chord covers of a fixed graph with a spanning 2-factor, nonzero scalar checks have rate tending to zero; the result extends uniformly to arbitrary-width blocks preserving a common full flag and having invertible edge maps. An exact homogenization construction preserves rate but can produce one large absolutely simple module, so separation does not imply the small-factor hypothesis. The scalar cover bounds also control ordinary group-word recovery over arbitrarily growing finite p-group alphabets when the exponent matrices modulo p are homogeneous.

Separately, over characteristic two, let $\mathcal{L}_R(\mathcal{C})$ be the span of codewords supported in radius-R balls. If the recovery graph has no odd cycle of length at most $2R+1$, then $\dim \mathcal{C} + \dim \mathcal{L}_R(\mathcal{C}) \leq \sum_v \dim \pi_v(\mathcal{C})$. For nonlinear encoders whose entire affine hull is recoverable, this implies a support-size and entropy payment for sources with nonlocal influence; all-local sources force the hull itself to have half rate. Exact homogenization and the kernel–range, quotient, and split-generator dimension interfaces are verified in Lean. The asymptotic cover theorems and the graph-local and source/group-word applications have complete written proofs but are not formalized. Neither a general nonlinear coding bound nor a native tape realization is supplied.

Contributions. Hans Brende chose the problem and directed the research. GPT-6 Astra (OpenAI) developed the arguments, text, and Lean formalizations under his direction. The imported rank-approximation theorem and other established inputs are attributed separately.

Contents

1 Introduction **3**
1.1 The results . . . 3
1.2 Relation to existing work . . . 4
1.3 Connection to transposition . . . 4

2 Templates, quotients, and checks **5**

3 Finite separation and rank transfer **6**

4 Uniformity over coefficient fields **8**

5 Growing widths and coefficient modules **9**

6 Nonzero checks on independent-chord covers **11**

7 Ordinary group-word recovery **13**

8 Exact homogenization and its width obstruction **14**

9 Local generation in characteristic two **15**
9.1 Nonlinear sources and a recoverable affine hull . . . 17

10 Construction requirements and the next questions **19**
10.1 A focused constructive target . . . 19
10.2 The corresponding impossibility targets . . . 19

A Index of Lean declarations **20**

B Proof status and verification **20**

1 Introduction

A recovery graph specifies which coordinates may determine another coordinate. It does not specify how a code is represented or computed. This distinction matters in the coding approach to matrix transposition: a structural property of a machine's dependency graph becomes useful only when a coding theorem applies to its actual code fibres. The companion paper [5] obtains long matched-port distances from native tape geometry and studies several coding bounds with additional premises. Here we examine two such premises more closely: algebraic structure in regular covers, and local generation of a linear code.

In the first setting a fixed finite template is repeated over the sheets of finite group quotients. Recovery matrices use the same coefficients at each sheet. We prove a half-rate theorem that is uniform over the coefficient field and values at fixed width. We then identify exactly what a filtration argument needs when the width grows. In the second setting the graph need not be a cover at all. Locally supported codewords are forced into the image of a check operator, while the whole code lies in its kernel. Rank-nullity gives a quantitative restriction on their span.

1.1 The results

The principal conclusions are the following.

(1) **Uniform half rate on fixed covers.** Under the precise ambient-group and chain hypotheses in Section 2, paired-port disconnection in the infinite incidence cover forces asymptotic rate at most $1/2$ for homogeneous linear checks of fixed total width. One threshold works for every coefficient tuple over every field of the fixed characteristic (Theorem 4.3). The coefficients and fields may therefore vary along the sequence.

(2) **An obstruction to excess rate at growing width.** Let $\alpha_i(k)$ be the fraction of internal dimension in simple coefficient-module factors larger than k, after algebraic closure. For each fixed k and $\varepsilon > 0$, eventually

$$r_i \le (1 - \alpha_i(k))(1/2 + \varepsilon) + \alpha_i(k).$$

Thus a sustained excess δ over half rate requires at least 2δ of the dimension in factors exceeding every fixed cutoff (Theorem 5.1 and Corollary 5.2).

(3) **Uniform zero rate for a stricter check class.** On the independent-chord covers of a fixed graph with a spanning 2-factor, every homogeneous scalar check with nonzero edge weights has rate tending to zero. The convergence is uniform in the field, weights, and diagonal entries. It also holds for arbitrary-width blocks with a common full flag and invertible edge maps (Theorem 6.2 and Corollary 6.3).

(4) **The cost of removing sheet dependence.** Any sheet-dependent linear check has an exact homogeneous inflation with the same rate. Its internal width grows by the sheet count, and its coefficient algebra can be a full matrix algebra of that growing size. Hence this conversion does not supply the small-factor premise (Propositions 8.1 and 8.2).

(5) **A local-generation deficit.** In characteristic two and with no odd cycle of length at most $2R + 1$,

$$\dim \mathcal{C} + \dim \mathcal{L}_R(\mathcal{C}) \le \sum_v \dim \pi_v(\mathcal{C}).$$

An above-half code must therefore have substantial dimension outside the span of its locally supported words (Theorem 9.2). If a nonlinear Cartesian-source encoder has a recoverable

affine hull, nonlocal sources must pay for excess support size or entropy (Theorem 9.4). All-local sources generate a hull of at most half rate.

(6) **Group-word extensions of the cover bounds.** Ordinary-word recovery over finite p-groups reduces to the scalar exponent check modulo p. Homogeneity of that check transfers the fixed-width half bound, and nonzero homogeneous edge exponents transfer the independent-chord zero-rate bound. The alphabet groups may grow without affecting the threshold (Corollary 7.2).

1.2 Relation to existing work

Storage codes and index coding are established subjects. In particular, triangle-free graphs can support binary linear codes with rate approaching one [2, 1, 11]. We make no general half-rate claim for triangle-free graphs. Our local theorem instead constrains the span of local codewords on such graphs; recovery locality and generator locality are different conditions.

The cover argument uses standard free-group subgroup separability, presented here by finite Schreier-graph completion; see Stallings [9] for the classical graph method. The free-group algebra input is Cohn's Sylvester-domain theorem [7], as recorded in [8, Introduction]. The asymptotic input is precisely Jaikin-Zapirain and Souza's Theorem 8.1 [8]. We apply that theorem to the specified quotients, then prove a separate uniformization over coefficient fields and parameters. We do not claim a new general rank-approximation theorem. The parameter argument uses the finite-residue-field consequence of the weak Nullstellensatz [10].

The first paper formulates the tape problem and its coding program [4]; the subsequent local-barrier paper separates cheap encoding from cheap local recovery [6]. The fourth paper supplies depth reduction and the reduction from a correct trace to a storage-code fibre [3]. The directed-recovery companion proves native paired-distance bounds and an all-width matrix-label criterion on a different family [5]. Our cover theorems neither subsume that all-width criterion nor establish its hypotheses for arbitrary cones. They remove field and coefficient restrictions in a specified cover model and expose the remaining width obstruction.

Proof status and attribution. Exact homogenization and the dimension interfaces, including the quotient deficit and split-generator criterion, are Lean-verified. Their formal statements are authoritative; the English outlines are guides to them. The asymptotic cover results, graph-local argument, and source and group-word applications have complete written proofs but are not formalized. In particular, the split-generator interface does not itself verify the graph parity and boundary argument. Every numbered result states its status. We distinguish the applications developed here from their standard algebraic inputs; external priority for the resulting combinations has not been established.

1.3 Connection to transposition

Keep the series' notation: d is the matrix side length, $b(d) = \max\{1, \lceil \log_2 d \rceil\}$, and $L = d^2 b(d)$. Statement (T) asks for one fixed multitape machine transposing these inputs in $O(L)$ time. A coding lower-bound route needs a code deficit that exceeds a paid genie cost. The present theorems can supply such a deficit only after their representation and linearity hypotheses have been proved for the relevant fibres. They are also tests for constructions: an above-half candidate in these regimes must escape the stated module or local-generation restrictions. The missing native implications are listed in Section 10; (T) remains open.

2 Templates, quotients, and checks

Fix a prime p and a finitely generated abstract free group Γ. A finite directed voltage template consists of a finite nonempty set V of types and a finite set E of triples $e = (u, v, g)$ with $u, v \in V$ and $g \in \Gamma$. The triple permits a type-u input for a type-v receiver. Equal triples are collected before assigning coefficients. Words that represent different elements of Γ remain separate even if they later coincide in a quotient.

For a quotient $q : \Gamma \twoheadrightarrow Q$, the incidence lift I_Q has ports $L_{(v,x)}, R_{(v,x)}$ for $v \in V, x \in Q$, with edges

$$L_{(u,x)} \; -\!- \; R_{(v,q(g)x)} \qquad ((u, v, g) \in E). \tag{2.1}$$

This graph is undirected for connectivity and distance; the recovery relation itself need not be symmetric. The two ports of (v, x) are *matched*. The infinite lift I_Γ uses the identity quotient. Common right multiplication of sheets is a graph automorphism. Our separation hypothesis is

$$L_{(v,1)} \text{ and } R_{(v,1)} \text{ lie in different components of } I_\Gamma \quad \text{for every } v \in V. \tag{2.2}$$

It then holds at every sheet. The literal template is retained even when selected coefficients vanish or cancel.

Give type v a finite-dimensional space W_v over a field K, put $w_v = \dim_K W_v$ and $B = \sum_v w_v > 0$, and choose $C_e : W_u \to W_v$ for every $e = (u, v, g)$. Let T_e be its extension by zero to $W = \bigoplus_v W_v$. With $\rho_Q(g)e_x = e_{q(g)x}$, define the homogeneous check

$$M_Q = I - \sum_{e \in E} T_e \otimes \rho_Q(g_e), \qquad (M_Q x)_{v,y} = x_{v,y} - \sum_{e=(u,v,g)} C_e x_{u,q(g)^{-1}y}. \tag{2.3}$$

Its normalized nullity is $r_Q = \dim_K \ker M_Q/(B|Q|)$. Every linear subcode of this kernel has at most this rate. Over finite K, the kernel rate is $\log_{|K|} |\ker M_Q|/(B|Q|)$. Homogeneity means that C_e does not depend on the sheet y.

A linear storage space with deterministic recovery admits linear recovery maps: on the attained parent projection define the map taking a parent tuple to its determined coordinate, then extend it linearly by a basis. This observation does not force homogeneous choices when different sheets originally have different maps. Homogeneity is a separate hypothesis of the cover theorems.

The actual chain. For the rank limits, fix a faithful embedding of Γ in a free-by-$\mathbb{Z}_p$ pro-p group $\mathcal{G}$. This means that $\mathcal{G}$ has a closed normal free pro-p subgroup with quotient $\mathbb{Z}_p$. Fix descending normal open subgroups U_i with $\bigcap_i U_i = \{1\}$ in $\mathcal{G}$, and put

$$H_i = \Gamma \cap U_i, \qquad Q_i = \Gamma/H_i, \qquad q_i = |Q_i|. \tag{2.4}$$

The normalization is q_i, not $|\mathcal{G}/U_i|$ unless the two quotients coincide. No cofinality in the separate pro-p completion of Γ is assumed. The natural maps $Q_j \to Q_i$, $j \geq i$, are surjective and have p-group kernels.

The prime, template, embedding, and chain stay fixed in every cover limit below. Widths are fixed through Section 4; thereafter each growing-width statement specifies its hypotheses. Zero-width types may be omitted. When nonidentity within-type voltages occur, an early quotient may collapse one to 1; then the matrix is still a check but need not describe recovery without self-input. Separation excludes identity within-type voltages, and the finite template and trivial intersection ensure that no such collapse occurs at sufficiently late levels.

One filtration fact used throughout. If a linear map T preserves a finite filtration $0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m = X$, and T_j is its map on F_j/F_{j-1}, then

$$\dim \ker T \le \sum_{j=1}^{m} \dim \ker T_j. \tag{2.5}$$

Indeed the map from $(\ker T \cap F_j)/(\ker T \cap F_{j-1})$ to F_j/F_{j-1} is injective and lands in $\ker T_j$. Sum dimensions. The filtration need not split, and quotient kernel vectors need not lift.

3 Finite separation and rank transfer

Lemma 3.1 (One finite separator)**.** Not formalized; a complete proof is provided.

Under (2.2), there is a finite quotient $q : \Gamma \twoheadrightarrow Q$ in which every matched port pair is still disconnected. This quotient need not be a p-group or a member of the chain (2.4).

Proof (not formalized). The finite base incidence multigraph has ports L_v, R_v and one edge per triple. Orient a traversal from L_u to R_v with label g and its reverse with label g^{-1}. The voltage of a path with chronological labels $a_1, \ldots, a_m$ is $a_m \cdots a_1$.

In a base component D, choose a root and a spanning tree. Let t_z be the voltage of the root-to-z tree path and let J_D be the subgroup of voltages of closed root walks. It is generated by the finitely many elements $t_y^{-1} a t_x$ from oriented edges $x \xrightarrow{a} y$: writing $a = t_y(t_y^{-1} a t_x)t_x^{-1}$ makes the tree labels telescope along a closed walk. The path voltages from x to y are exactly $t_y J_D t_x^{-1}$, by closing a path at the root, and conversely inserting a closed root walk.

If L_v, R_v are in different base components, their lifts are always disconnected. Otherwise their matched lifts in a quotient are connected exactly when

$$q(t_{R_v}^{-1} t_{L_v}) \in q(J_D).$$

Thus (2.2) gives a finite list of pairs (J, g) with J finitely generated and $g \notin J$.

Here is the required finite separation argument. In the coset graph with vertices zJ, retain the paths from J spelled by a finite generating set of J and by g. For a left action, traverse a word from its rightmost letter. Their finite vertex union S contains both J and the distinct vertex gJ. Each free generator acts on the retained edges by a partial injection of S into itself. Complete it to a permutation, matching the unused domain and range vertices. Freeness extends these permutations to a homomorphism $\Gamma \to \mathrm{Sym}(S)$. The retained generator paths of J fix the vertex J, while g does not. The image of g is therefore outside the image of J.

Take the product of these finitely many separating homomorphisms and let Q be its image. Each forbidden membership fails in its designated factor, hence in Q. If the list is empty, use the trivial quotient. □

This is the standard subgroup-separation mechanism specialized to the two ports of every type. Its role is to supply a finite rank witness; it does not replace the actual chain by this auxiliary quotient.

Proposition 3.2 (A finite rank witness)**.** Not formalized; a complete proof is provided.

For any field K and widths of total $B > 0$, let $M = I - \sum_e T_e g_e \in \mathrm{Mat}_B(K[\Gamma])$ fit a template satisfying (2.2). Then

$$\mathrm{irk}_{K[\Gamma]} M \ge \lceil B/2 \rceil.$$

Here inner rank is the minimum k for a factorization through k columns over the group algebra.

Proof (not formalized). Use the quotient Q from Lemma 3.1. Order its incidence components uniformly at random. Select a coordinate (v, x) when its left component precedes its right component, and give it weight w_v. The components differ, so its selection probability is $1/2$. Some order selects total weight at least $B|Q|/2$.

For a selected input α and selected receiver β joined by a dependency, the left component of α equals the right component of β. Selection of β therefore places its left component strictly before that of α. The selected principal block of M_Q, ordered by left component, is triangular with identity diagonal. Consequently $\operatorname{rank}_K M_Q \geq B|Q|/2$.

A factorization $M = XY$ through k columns specializes under the regular representation to a factorization of M_Q through $k|Q|$ columns. Hence $k|Q| \geq B|Q|/2$ and $k \geq \lceil B/2 \rceil$. No semisimplicity is needed, even when p divides $|Q|$. □

Published algebraic inputs. Cohn's theorem identifies inner rank over a free-group algebra with rank in its universal division ring [7, 8]. For the subgroup and chain in (2.4), Jaikin-Zapirain–Souza [8, Theorem 8.1] gives, for each fixed matrix M over $\mathbb{F}_p[\Gamma]$,

$$\lim_i \frac{\operatorname{rank}_{\mathbb{F}_p} M_{Q_i}}{q_i} = \operatorname{irk}_{\mathbb{F}_p[\Gamma]} M. \tag{3.1}$$

These are imported theorems, not formalized here. The source uses a right action on rows. Its finite rank equals our left-action rank on columns: the pairing $\langle y, x \rangle = \tau(\sum_j y_j x_j)$, where τ extracts the identity coefficient in $\mathbb{F}_p[Q_i]$, is nondegenerate and satisfies $\langle y, Mx \rangle = \langle yM, x \rangle$.

Theorem 3.3 (Fixed prime-field checks)**.** Not formalized; a complete proof is provided.
With (2.4) and (2.2), every fixed $\mathbb{F}_p$ check of total width $B > 0$ satisfies

$$\lim_i \frac{\dim_{\mathbb{F}_p} \ker M_{Q_i}}{Bq_i} \leq \frac{\lfloor B/2 \rfloor}{B}.$$

The same upper bound on the limsup holds when the $\mathbb{F}_p$ coefficients vary with i, while the template and widths remain fixed.

Proof (not formalized). Combine Proposition 3.2 with (3.1) and rank-nullity. There are finitely many $\mathbb{F}_p$ assignments to the fixed coefficient entries, so the maximum of their convergence thresholds works for every assignment at once. □

For a fixed finite extension $K/\mathbb{F}_p$ of degree a, restriction of scalars multiplies both kernel dimension and B by a. Thus the pointwise half bound holds over K, with the possibly stronger bound $\lfloor aB/2 \rfloor/(aB)$. This argument alone does not yield $\lfloor B/2 \rfloor/B$ over K or a uniform threshold over all a. The next section supplies the latter uniformity at the half constant.

Lemma 3.4 (Long paired distances imply infinite separation)**.** Not formalized; a complete proof is provided.
For a fixed template with $n = |V| > 0$, suppose finite quotient lifts satisfy, for every fixed integer $r \geq 0$,

$$\#\{(v, x) : \operatorname{dist}_{I_{Q_i}}(L_{(v,x)}, R_{(v,x)}) \leq r\} = o(nq_i).$$

Then (2.2) holds. This implication needs no nesting or prime-power condition on the quotients.

Proof (not formalized). An infinite-cover matched path at one type has finite length r. It descends to a walk of that length in every quotient. Right translation supplies such a walk at all q_i sheets of that type, giving proportion at least $1/n$, a contradiction. □

Thus the native paired-distance invariant can supply the separation premise *if* a fixed-template cover representation is already available. The rank proof provides no such representation.

4 Uniformity over coefficient fields

The pointwise rank limit does not by itself allow the field or coefficients to change at each level. We prove the required uniformity in two steps. This section first works with any group Γ and compatible finite quotients Q_i, provided every refinement kernel is a p-group.

Lemma 4.1 (p-refinement monotonicity)**.** Not formalized; a complete proof is provided.

Let K have characteristic p, and fix a square matrix $M \in \mathrm{Mat}_B(K[\Gamma])$ with $B > 0$. If $Q_j \twoheadrightarrow Q_i$ has kernel P, a finite p-group, then

$$\dim_K \ker M_{Q_j} \le |P| \dim_K \ker M_{Q_i}, \qquad \frac{\dim_K \ker M_{Q_j}}{Bq_j} \le \frac{\dim_K \ker M_{Q_i}}{Bq_i}.$$

The field and coefficients are fixed within this comparison.

Proof (not formalized). Choose a section $s : Q_i \to Q_j$. Each fine sheet is uniquely $s(x)k$ with $k \in P$, and left multiplication by g takes it to

$$s(g_i x)\kappa(g, x)k, \qquad \kappa(g, x) = s(g_i x)^{-1} g_j s(x) \in P.$$

If $M_{ab} = \sum_g c_{ab,g} g$, its fine block from coarse column (b, x) to coarse row (a, y) is the operator

$$\sum_{g : g_i x = y} c_{ab,g} \rho_P(\kappa(g, x)) \quad \text{on } K[P].$$

Every nonzero finite-dimensional $K[P]$-module has a nonzero fixed vector. Induct on $|P|$: a nontrivial finite p-group has a central element z of order p (by its class equation and Cauchy's theorem), and $(z - I)^p = 0$. Its nonzero kernel is P-invariant, with action factoring through $P/\langle z \rangle$; induction applies there. For trivial P there is nothing to prove. Repeating on quotients gives a full flag in $K[P]$ with one-dimensional trivial factors.

Tensor this flag with the coarse coordinate space. On every quotient layer, each $\rho_P(\kappa)$ acts as 1, so the induced matrix is exactly M_{Q_i}. All colliding words contribute their scalar sum; none is discarded. Apply (2.5) to the $|P|$ layers and use $q_j = |P| q_i$. □

Theorem 4.2 (Finite-type uniformization)**.** Not formalized; a complete proof is provided.

Let $\mathscr{R}$ be a finite-type commutative unital $\mathbb{F}_p$-algebra, and let $M \in \mathrm{Mat}_B(\mathscr{R}[\Gamma])$, $B > 0$, have fixed finite support. Use compatible finite quotients with p-group refinement kernels. For a unital map $a : \mathscr{R} \to K$ to a field, write

$$r_i(a) = \frac{\dim_K \ker M_{Q_i}(a)}{Bq_i}.$$

Suppose $0 \le \beta < 1$ and $\limsup_i r_i(a) \le \beta$ for every such map to a finite *field. For every $\varepsilon > 0$ there is I such that*

$$r_i(a) < \beta + \varepsilon$$

for all $i \ge I$, all characteristic-p fields K, and all unital maps $a : \mathscr{R} \to K$.

Proof (not formalized). The zero ring has no unital maps to fields. If $\theta = \beta + \varepsilon > 1$, the assertion follows from $r_i \le 1$. Otherwise $0 < \theta \le 1$. Set $N_i = Bq_i$, $k_i = \lfloor (1-\theta)N_i \rfloor$, and let J_i be the ideal of $\mathscr{R}$ generated by all (k_i+1)-minors of M_{Q_i}. For every field specialization,

$$r_i(a) \ge \theta \quad \Longleftrightarrow \quad \operatorname{rank} M_{Q_i}(a) \le k_i \quad \Longleftrightarrow \quad a(J_i) = 0. \tag{4.1}$$

The rounding is exact, including $\theta = 1$.

If $J = \sum_i J_i$ were proper, choose a maximal ideal $\mathfrak{m} \supseteq J$. By the finite-type weak Nullstellensatz [10, Theorem 10.34.1], $\mathscr{R}/\mathfrak{m}$ is a finite extension of $\mathbb{F}_p$. Its residue map would be bad at every level by (4.1), contradicting the pointwise hypothesis. Thus $J = \mathscr{R}$ and there is a finite identity

$$1 = \sum_{\ell=1}^{m} f_\ell \Delta_\ell, \tag{4.2}$$

where Δ_ℓ is a generating minor at some level i_ℓ. Take $I = \max_\ell i_\ell$.

If a specialization were bad at $j \ge I$, then Lemma 4.1, applied to that same field and tuple, would make it bad at every i_ℓ. All $a(\Delta_\ell)$ would vanish, contradicting (4.2). This proves the claim. The ideals J_i need not be nested; only their bad field loci are ordered by monotonicity. □

This argument also allows equations among parameters and finitely many nonzero constraints, represented by inverse variables. It supplies an eventual threshold, not a useful numerical estimate for it. A search for (4.2) is possible when the ring, chain, and matrices are effectively presented and the threshold is rational; no running-time bound is asserted.

Theorem 4.3 (Field-uniform half rate)**.** Not formalized; a complete proof is provided.

Fix the template, embedding, and chain of Section 2, and suppose (2.2). Fix widths $w_v \ge 0$ of total $B > 0$. For every $\varepsilon > 0$ there is I such that for every $i \ge I$, every characteristic-p field K, and every homogeneous coefficient tuple fitting the template,

$$\frac{\dim_K \ker M_{Q_i}}{Bq_i} < \frac{1}{2} + \varepsilon.$$

The threshold is independent of K and the coefficient tuple.

Proof (not formalized). Give each scalar entry of each C_e an independent variable and form the universal check over their polynomial ring over $\mathbb{F}_p$. Every finite-field tuple has pointwise rate at most $1/2$ by Theorem 3.3 and restriction of scalars. Remove zero-width types if necessary; the remaining incidence graph retains separation. Apply Theorem 4.2 with $\beta = 1/2$. □

This rules out above-half homogeneous constructions with fixed width even when they change fields and coefficients at every size. It does not bound arbitrary sheet-dependent maps or nonlinear code supports. The p-kernel condition is material: in characteristic three, $1 + a$ has nullity zero on the trivial quotient but positive nullity on C_2. Thus Lemma 4.1 is false for unrestricted refinements.

5 Growing widths and coefficient modules

Keep the fixed geometry and separation of Theorem 4.3, but at stage i allow an arbitrary field K_i of characteristic p and arbitrary spaces $W_{v,i}$, of total dimension $B_i > 0$. The homogeneous coefficient maps may also vary. Choose an algebraic closure Ω_i of K_i and set $\overline{W}_i = (\bigoplus_v W_{v,i}) \otimes_{K_i} \Omega_i$. Let $\mathcal{A}_i$ be the unital Ω_i-algebra generated by the vertex projectors and the embedded coefficient maps T_e.

Take a composition series of the $\mathcal{A}_i$-module $\overline{W}_i$. Write its simple-factor dimensions as $d_{i,1}, \dots, d_{i,m_i}$. For $k \geq 1$ put

$$\alpha_i(k) = \frac{1}{B_i} \sum_{j : d_{i,j} > k} d_{i,j}. \tag{5.1}$$

These are dimensions over Ω_i, not over $\mathbb{F}_p$. All statements below hold for any composition series. The Jordan–Hölder theorem also makes this multiset independent of the chosen series; extension of a field to its algebraic closure does not change a finite matrix's rank, since rank is detected by minors.

Theorem 5.1 (A bound from simple-factor mass)**.** Not formalized; a complete proof is provided.
For each fixed $k \geq 1$ and $0 < \varepsilon < 1/2$, one index $I(k, \varepsilon)$ works for all fields, widths, and homogeneous coefficient tuples: for all $i \geq I(k, \varepsilon)$,

$$r_i \leq (1 - \alpha_i(k))(1/2 + \varepsilon) + \alpha_i(k).$$

Proof (not formalized). Every invariant subspace N is graded by types: $N = \bigoplus_v E_v N$, because the vertex projectors E_v are in $\mathcal{A}_i$, are orthogonal, and sum to the identity. The grading passes to every invariant subquotient, and each T_e still maps its source piece to its target piece. Thus the induced check on a simple factor fits the same template, with some nonnegative vertex dimension vector of total $d_{i,j}$.

Tensor the composition series with $\Omega_i[Q_i]$ and apply (2.5). For the factors of total dimension at most k, there are only finitely many possible vertex dimension vectors. Take the maximum of the corresponding thresholds from Theorem 4.3. This maximum is independent of the field, coefficients, and number of factors. Every such factor contributes at most $(1/2 + \varepsilon) d_{i,j} q_i$ to nullity. Each larger factor contributes at most $d_{i,j} q_i$. Sum, divide by $B_i q_i$, and use $\sum_j d_{i,j} = B_i$. Field extension preserves the original rate. □

Corollary 5.2 (Tightness and excess)**.** Not formalized; a complete proof is provided.
If

$$\lim_{k \to \infty} \limsup_i \alpha_i(k) = 0,$$

then $\limsup_i r_i \leq 1/2$. *Conversely, if* $\liminf_i r_i \geq 1/2 + \delta$ *with* $0 < \delta \leq 1/2$, *then for every fixed k,*

$$\liminf_i \alpha_i(k) \geq 2\delta.$$

A limsup excess gives only the corresponding limsup conclusion.

Proof (not formalized). For fixed k, take limsup in Theorem 5.1 and then let $\varepsilon \downarrow 0$. This gives $\limsup r_i \leq 1/2 + (1/2) \limsup \alpha_i(k)$; now let $k \to \infty$. For the converse, rearrange the finite inequality as

$$\alpha_i(k) \geq \frac{r_i - 1/2 - \varepsilon}{1/2 - \varepsilon},$$

take liminf, and then let $\varepsilon \downarrow 0$. The same rearrangement with limsup proves the last assertion. No growing cutoff is substituted into a fixed-cutoff threshold. □

Corollary 5.3 (A common internal flag)**.** Not formalized; a complete proof is provided.
Suppose at each stage all $n = |V|$ vertex spaces are identified with one space Z_i of arbitrary positive dimension, and the internal maps C_e preserve a common full flag after algebraic closure. Then $\limsup_i r_i \leq 1/2$, *uniformly in fields and dimensions. Pairwise commutativity of the identified internal maps suffices for the flag hypothesis.*

Proof (not formalized). The direct sums over types of successive internal flag spaces are invariant under every projector and embedded coefficient map. Their successive quotients have total dimension n. Refining them to composition series gives $\alpha_i(n) = 0$; apply Theorem 5.1.

For commuting operators over an algebraically closed field, an eigenspace of one operator is invariant under the others. Successively restricting to eigenspaces gives a common eigenvector. Its invariant line and induction on the quotient give a full common flag. This proves simultaneous triangularizability, not diagonalizability. The embedded maps between different types need not commute. □

Why factors are taken after algebraic closure. There is a strict difference from measuring simple factors over the prime field. Take two types with identity maps in both directions and one extra forward map given by multiplication by a generator ζ of $\mathbb{F}_{p^m}/\mathbb{F}_p$. Use voltages $1, 1, a$ over $\Gamma = \langle a \rangle$ with quotients C_{p^i} from $\mathcal{G} = \mathbb{Z}_p \times \mathbb{Z}_p$ and $U_i = p^i\mathbb{Z}_p \times p^i\mathbb{Z}_p$. Paired ports are disconnected because the two incidence type classes are disjoint. Over $\mathbb{F}_p$ the coefficient algebra is $\mathrm{Mat}_2(\mathbb{F}_{p^m})$: the identity maps give the cross matrix units and the extra map generates all field scalars in a diagonal corner. Its natural module is simple of dimension $2m$. Over an algebraic closure the separable field extension splits into m copies of that field, so the module has m factors of dimension 2. Hence $\alpha_i(2) = 0$ even when m grows without bound. This is a comparison of hypotheses; bipartiteness already bounds every code on this particular relation by half rate.

The excess conclusion is a construction requirement, not a claim that large simple factors produce high-rate codes. The explicit example in Section 8 has maximal large-factor mass and nevertheless obeys an elementary all-code half bound.

6 Nonzero checks on independent-chord covers

For a stricter check class the limiting rate is zero. Fix a finite simple connected graph H on n vertices with a spanning 2-factor, that is, disjoint cycles covering all vertices. Choose a spanning tree and orient its chords. Let Γ be the free group on these chords. Assign reciprocal voltages $g_{uv} = g_{vu}^{-1}$ to adjacent ordered pairs, with tree voltages 1 and one independent generator on each oriented chord. Here g_{uv} is in matrix row u, column v; it takes a sheet at type v to the sheet multiplied on the left by g_{uv} at type u.

Fix an embedding and chain as in (2.4) for this abstract Γ. Over a characteristic-p field K, choose arbitrary diagonal scalars d_u and nonzero scalars c_{uv} for every ordered edge. Define

$$T_{uu} = d_u, \qquad T_{uv} = c_{uv}g_{uv} \quad (uv \in E(H)), \qquad T_{uv} = 0 \quad \text{otherwise.} \tag{6.1}$$

Reverse coefficients need not agree or be inverses.

Lemma 6.1 (A surviving determinant monomial)**.** Not formalized; a complete proof is provided.

Abelianize the chord generators to independent Laurent variables $z_1, \dots, z_s$. The resulting matrix T_z has nonzero determinant over $K[z_1^{\pm 1}, \dots, z_s^{\pm 1}]$ for every allowed tuple.

Proof (not formalized). Orient the cycles of a spanning 2-factor and let π_0 send each vertex to its successor. The associated determinant term has nonzero coefficient $\mathrm{sgn}(\pi_0) \prod_u c_{u,\pi_0(u)}$. For any determinant permutation, record the net directed use of each undirected edge. This integer current has divergence zero; the Laurent exponents are its chord coordinates, with fixed choices of sign. If a second permutation has the same exponents as π_0, the difference of the two currents is supported on the spanning tree and has divergence zero. Leaf deletion shows that it vanishes.

Every oriented factor edge has net use one. Since a permutation uses each ordered edge at most once, that current forces use of the same oriented edge. These edges prescribe one outgoing arrow at every vertex, so the permutation equals π_0. The selected monomial is unique and cannot cancel, in any characteristic. No factor vertex uses a diagonal, so arbitrary d_u cause no issue. □

Theorem 6.2 (Uniform scalar rate zero)**.** Not formalized; a complete proof is provided.
For every $\varepsilon > 0$ there is $I(\varepsilon)$ such that for all $i \geq I(\varepsilon)$, all characteristic-p fields, all nonzero directed edge weights, and all diagonal values in (6.1)*,*

$$\frac{\dim_K \ker T_{Q_i}}{nq_i} < \varepsilon.$$

Proof (not formalized). First fix one tuple over $K = \mathbb{F}_{p^a}$. Restrict scalars to $\mathbb{F}_p$ to obtain a fixed matrix T^{res} of size an over $\mathbb{F}_p[\Gamma]$. Its abelian specialization over $\mathbb{F}_p(z_1, \ldots, z_s)$ is invertible. To justify this step, an $\mathbb{F}_p$-basis of K remains linearly independent over $\mathbb{F}_p(z)$: clear denominators and compare monomial coefficients. Its $\mathbb{F}_p(z)$-span is a finite-dimensional subalgebra of the field $K(z)$, hence is a field, since multiplication by any nonzero element is injective and therefore surjective. It contains K and $\mathbb{F}_p(z)$, so it is all of $K(z)$. The specialization is consequently the restriction of scalars of the invertible map T_z from Lemma 6.1.

Any factorization of T^{res} through fewer than an columns would specialize to such a factorization of an invertible matrix over this rational-function field. Hence its inner rank is an. Equation (3.1) gives a limiting normalized nullity of zero. Restriction of scalars preserves that normalized nullity, so the fixed finite-field tuple has rate tending to zero.

Now form the finite-type parameter algebra

$$\mathscr{R} = \mathbb{F}_p[c_{uv}, c_{uv}^{-1} : uv \text{ an ordered edge}][d_u : u \in V(H)].$$

Its maps to fields are exactly the permitted tuples. Apply Theorem 4.2 to the universal matrix with $\beta = 0$. The nonzero constraints survive every residue-field specialization. □

Corollary 6.3 (Arbitrary widths with a common flag)**.** Not formalized; a complete proof is provided.
Replace the scalars in (6.1) *by internal matrices on one w_i-dimensional space at each type, over any characteristic-p field K_i. Assume that all edge and diagonal matrices preserve a common full flag after algebraic closure, and every edge matrix is invertible. Then the normalized nullity tends to zero uniformly in the fields, coefficients, flags, and widths $w_i \geq 1$.*

Proof (not formalized). On each one-dimensional internal flag quotient, the edge matrices induce nonzero scalars: an invertible triangular matrix has nonzero diagonal entries. The induced diagonal scalars are unrestricted. There are w_i scalar check layers, each on nq_i coordinates. The same threshold in Theorem 6.2 bounds every layer's nullity by εnq_i. Equation (2.5) then bounds the full nullity by εnw_iq_i. Extending fields preserves rank, so the estimate holds over K_i. □

Commuting internal edge and diagonal maps give the flag by the proof of Corollary 5.3; edge invertibility is still required. For a recovery check, choose diagonal I and edge coefficients equal to the negatives of the recovery maps. Thus this theorem excludes the corresponding full-neighbor, common-flag construction class, even at unrestricted vector width. It is a bound on specified checks, not the storage capacity of the underlying graphs.

The nonzero premise cannot be dropped. Set every diagonal to 1, retain only one edge uv with coefficients -1 in both directions, and set all other edge coefficients to zero. Its two-type block is

$$\begin{pmatrix} I & -P \\ -P^{-1} & I \end{pmatrix}, \qquad P = \rho_{Q_i}(g_{uv}),$$

which has kernel dimension q_i, while all other types have identity checks. Its normalized nullity is $1/n$ at every level. No paired-distance hypothesis was needed for the rate-zero theorem; the independent chord variables and nonzero weights supplied the stronger rank certificate directly.

7 Ordinary group-word recovery

The scalar cover bounds also apply to a nonlinear class with arbitrarily large finite alphabets. An ordinary group word uses variables, their inverses, multiplication, and fixed constants. The expression at a coordinate is fixed on the whole code; arbitrary symbol maps and endomorphisms are not included here. The latter require the vector-layer benchmark of [5, Theorem B.5].

Lemma 7.1 (The scalar exponent check)**.** Not formalized; a complete proof is provided.

Let G be a finite p-group and $\mathcal{D} \subseteq G^N$ a set satisfying fixed ordinary-word equations $x_v = w_v(x)$. Let $E_{vu} \in \mathbb{F}_p$ be the signed exponent sum of variable x_u in w_v, and put $k = \dim_{\mathbb{F}_p} \ker(I - E)$. Then

$$|\mathcal{D}| \le |G|^k.$$

For any law on a nonempty $\mathcal{D}$, its entropy is at most $k \log_2 |G|$.

Proof (not formalized). Use the terminating lower exponent-p central series $P_1 = G$, $P_{j+1} = P_j^p[P_j, G]$. Its termination and the central-layer argument are proved in [5, Theorem B.5]. Write $t_j = \dim_{\mathbb{F}_p}(P_j/P_{j+1})$, so $\sum_j t_j = \log_p |G|$. Fix an attained solution modulo P_j and choose representatives modulo P_{j+1}. Each further lift has the form $r_v a_v$, where a_v is in the central elementary abelian layer. Centrality makes the change in an ordinary word equal to $\sum_u E_{vu} a_u$: products add, inversion negates, and constants give zero. Thus the lift equations have linear part $(I - E) \otimes I_{t_j}$. A nonempty fibre is contained in an affine solution space of size p^{kt_j}. Multiplying the fibre bounds gives $|\mathcal{D}| \le p^{k \sum_j t_j} = |G|^k$. Empty sets and trivial groups satisfy the cardinality bound directly. Entropy is at most the logarithm of the support size. □

This is the ordinary-word specialization of the classical central lifting method discussed in [5], not a new general group-equation theorem. In particular, a zero exponent sum is a statement about this word language, not a rule for erasing arbitrary nonlinear operations from a recovery circuit.

Corollary 7.2 (Uniform bounds for growing group alphabets)**.** Not formalized; a complete proof is provided.

Fix the template, prime, embedding, chain, and positive integer type widths of Theorem 4.3. At level i, use any nontrivial finite p-group G_i as the alphabet in each of the $N_i = Bq_i$ symbol positions. Let a nonempty code $\mathcal{D}_i \subseteq G_i^{N_i}$ have ordinary-word recovery from the permitted parents. Suppose its exponent matrix modulo p is homogeneous on the external sheets, with the block shapes of the fixed template. For every $\varepsilon > 0$, eventually

$$\frac{\log_{|G_i|} |\mathcal{D}_i|}{N_i} < \frac{1}{2} + \varepsilon,$$

with one threshold independent of G_i, its order, the words, and their constants.

On the independent-chord family of Theorem 6.2, use one G_i-symbol per vertex. If the exponent on each directed edge is homogeneous and nonzero modulo p, the same normalized rate tends to zero, uniformly in these choices. The corresponding entropy bounds hold for every law on each code.

Proof (not formalized). The exponent check $I - E_i$ is a characteristic-p homogeneous linear check with the stated fixed type widths. Apply Lemma 7.1 to bound the normalized logarithmic size by $\dim_{\mathbb{F}_p} \ker(I - E_i)/N_i$, then apply Theorem 4.3. For the second conclusion, the diagonal is 1 and the directed edge coefficients are the negatives of the nonzero exponent sums. They meet Theorem 6.2 exactly. Its uniform threshold gives the zero-rate conclusion. The entropy assertions follow from the final part of Lemma 7.1. □

The words themselves may vary between sheets: only their exponent matrices must be homogeneous. Reordering occurrences, changing constants, and inserting commutators can preserve this matrix. The number of group-symbol slots per type remains fixed, even though the group alphabet may grow. Arbitrary endomorphic expressions have matrix-valued central derivatives; the scalar argument above does not cover them. Nor does it supply a group-word representation or a charged native implementation for an arbitrary tape fibre.

This extends the cover obstruction beyond linear codes while retaining an explicit algebraic restriction on recovery. An above-half candidate in this cover model must violate the word or exponent hypotheses, or the cover hypotheses themselves. The rate-zero conclusion additionally requires every prescribed edge exponent to be nonzero; the single-edge example of Section 6 shows why that condition matters.

8 Exact homogenization and its width obstruction

It is natural to try to encode sheet-dependent coefficients inside larger homogeneous blocks. The following transformation is exact, but its module structure explains why it does not settle the unrestricted-width problem.

Proposition 8.1 (Rate-preserving inflation)**.** Lean: `sheetCheck`, `inflatedCheck`, `sheetReindex`, `reindex_inflated`, `kernelEquiv`, `finrank_kernel_inflated`, `rate_cross_product` [`HomogeneousInflation.lean`]

On a finite quotient Q of order q, allow coefficients indexed by the destination sheet:

$$(M_C x)_{v,t} = x_{v,t} - \sum_{e=(u,v,g)} C_e(t) x_{u,g^{-1}t}.$$

There is a homogeneous check on the same external voltage support, with each internal width multiplied by q, that is permutation equivalent to q copies of M_C. Its normalized nullity is unchanged.

Outline of the formal proof. The formalization takes $W = \bigoplus_v W_v$ and extends every $C_e(t)$ by zero from its source type to its target type. Thus one sheet is a vector in W, and the two checks below are exactly `sheetCheck` and `inflatedCheck` on W-valued functions. Replace W_v by $\widetilde{W}_v = \bigoplus_{t \in Q} W_v$ and set $(D_e z)_t = C_e(t) z_{g^{-1}t}$. The homogeneous check is $\widetilde{M} = I - \sum_e D_e \otimes \rho_Q(g_e)$. Its coordinates $x_{v,x,t}$ have external sheet x and internal sheet t, and a dependency simultaneously replaces these by $g^{-1}x, g^{-1}t$. Therefore $h = x^{-1}t$ is invariant. For each $h \in Q$ put $y^h_{v,t} = x_{v,th^{-1},t}$. This bijective reindexing turns the check into the original equations for y^h, independently for each h.

The formal kernel equivalence is $\ker \widetilde{M} \cong \prod_{h \in Q} \ker M_C$. Mathlib's dimension formulas therefore give $\dim \ker \widetilde{M} = q \dim \ker M_C$, while the ambient dimensions are respectively Bq^2 and Bq. The Lean rate identity is stated by cross multiplication, so it also holds at zero width; dividing by the positive ambient dimensions gives the stated rate equality. Neither commutativity of Q nor distinct voltages is assumed. □

For a linear subcode $\mathcal{C} \leq \ker M_C$, the same coordinate permutation takes $\prod_{h\in Q} \mathcal{C}$ into the inflated kernel. Its dimension is $q \dim \mathcal{C}$, so it too preserves rate.

Proposition 8.2 (One large absolutely simple factor)**.** Not formalized; a complete proof is provided. *There is a fixed paired-separated template on an actual chain (2.4) for which the inflation in Proposition 8.1 produces a coefficient module that remains simple after every field extension, of dimension $2q_i \to \infty$. Nevertheless every storage code on the underlying two-type relation has rate at most one half for equal coordinate alphabets.*

Proof (not formalized). Take $\Gamma = \langle a \rangle \subset \mathcal{G} = \mathbb{Z}_p \times \mathbb{Z}_p$ by $a \mapsto (1, 0)$, with $U_i = p^i\mathbb{Z}_p \times p^i\mathbb{Z}_p$. Then $Q_i = C_{p^i}$. Use two types $0, 1$ and four triples

$$(0, 1, 1), \quad (1, 0, 1), \quad (0, 1, a), \quad (0, 1, a^2).$$

Their incidence edges stay in the disjoint classes $\{L_0, R_1\}$ and $\{L_1, R_0\}$, so every matched pair is disconnected in the infinite and finite lifts.

For scalar coordinates over $\mathbb{F}_p$, choose destination-sheet coefficients $1, 1, 1, \mathbf{1}_{t=1}$ in that order. On the inflated internal space $Z_i = \mathbb{F}_p^{Q_i}$, let $Pe_t = e_{at}$ and let F project onto the identity sheet. The four internal maps are I, I, P, FP^2. Include the two vertex projectors in their algebra. The cross identities put P and FP^2 in one diagonal corner. Since $P^{q_i} = I$, that corner also contains P^{-2} and therefore F. For any sheets x, y, powers of P give $P_xFP_y^{-1} = E_{xy}$, the corresponding matrix unit. The cross identities transfer these units to all four type corners. Consequently the algebra is $\mathrm{Mat}_{2q_i}(\mathbb{F}_p)$, and remains the full matrix algebra after any field extension. Its natural module is simple: a full endomorphism algebra can send any nonzero vector to any specified vector. Thus $\alpha_i(k) = 1$ eventually for every fixed k.

Finally each type is recoverable from the other type alone. Projection onto either full type is injective on any storage code. With q_i coordinates per type and common alphabet Σ, $|\mathcal{C}| \leq |\Sigma|^{q_i}$. The same argument applies to the inflated block alphabet. No rate attainment by the displayed check is asserted. □

The example refutes automatic small-factor mass after homogenization. It does not refute a possible all-width half-rate theorem. In particular, the contrapositive of Corollary 5.2 provides a necessary algebraic escape condition, not a sufficient construction.

9 Local generation in characteristic two

This section is independent of the group-cover hypotheses. Let G be a finite simple undirected graph, let K be a characteristic-two field, let each W_v be finite-dimensional, and let $\mathcal{C} \leq \bigoplus_{v\in V(G)} W_v$ be a linear code. For each v, a specified set $N_v \subseteq N_G(v)$ determines x_v on the code. The supplied recovery functions may be nonlinear; only the code support is assumed linear. Write $U_v = \pi_v(\mathcal{C})$ and $W = \bigoplus_v U_v$ for the attained coordinate spaces. Let

$$\mathcal{L}_R(\mathcal{C}) = \mathrm{span}_K\{x \in \mathcal{C} : \mathrm{supp}\, x \subseteq B_R(c) \text{ for some } c \in V(G)\}.$$

Here $B_R(c)$ is the closed graph-distance ball of radius R about c. The center may depend on the codeword. This is a condition on generators of the actual code, not on its recovery equations.

Lemma 9.1 (The dimension interface)**.** Lean: `kernel_range_dimension_bound`, `quotient_dimension_bound`, `split_generator_dimension_bound` [`LocalGeneration.lean`]

Let W be a finite-dimensional vector space over a division ring, $H : W \to W$ linear, and C, L subspaces with $C \subseteq \ker H$ and $L \subseteq \operatorname{im} H$. Then $\dim C + \dim L \le \dim W$. If also $L \subseteq C$, then

$$2 \dim C \le \dim(C/L) + \dim W.$$

For $H = I + A$ and $L = \operatorname{span} S$, the range premise follows whenever every $x \in S$ admits $x = s + As$ for some $s \in W$.

Outline of the formal proof. Monotonicity of dimension bounds the two summands by $\dim \ker H$ and $\dim \operatorname{im} H$. Mathlib's rank-nullity theorem says their sum is $\dim W$. Mathlib's quotient dimension formula gives the second inequality. For the last assertion, $x = (I + A)s$ puts each generator in the range; closure under linear combinations puts the span there as well. The paper library reuses these shared lemmas; no graph premise is hidden in their formal statements. □

Theorem 9.2 (The local span cannot exceed the deficiency)**.** Not formalized; a complete proof is provided.

If G has no odd cycle of length at most $2R + 1$, where $R \ge 0$ is an integer, then

$$\dim_K \mathcal{C} + \dim_K \mathcal{L}_R(\mathcal{C}) \le \sum_v \dim_K U_v. \tag{9.1}$$

Consequently local generation, $\mathcal{L}_R(\mathcal{C}) = \mathcal{C}$, implies the weighted half bound. More generally,

$$\dim_K(\mathcal{C}/\mathcal{L}_R(\mathcal{C})) \ge 2 \dim_K \mathcal{C} - \sum_v \dim_K U_v.$$

No degree, common-width, or reciprocal-recovery hypothesis is required.

Proof (not formalized). For each v, map an attained parent tuple to its determined coordinate in U_v. This map is well-defined by recovery and linear because both projections come from the linear space $\mathcal{C}$. Extend it from the attained parent subspace to $\bigoplus_{u \in N_v} U_u$ by extending a basis. We obtain fixed linear maps $A_{vu} : U_u \to U_v$ such that every codeword satisfies $x_v = \sum_{u \in N_v} A_{vu} x_u$. Let A be this ambient recovery operator on W, and put $H = I + A$. Characteristic two gives $\mathcal{C} \subseteq \ker H$.

Take a codeword x supported in $B_R(c)$. Color vertices of that ball by the parity of their distance from c. An edge within the ball whose endpoints had the same parity would, together with shortest paths to c, give an odd closed walk of length at most $2R + 1$. Removing repeated-vertex subwalks leaves an odd cycle no longer than that walk, contrary to the hypothesis. Thus every edge inside the ball joins opposite parities.

Split $x = s + t$ by these two parities, extending each part by zero outside the ball. We claim $As = t$ and $At = s$. At a vertex inside the ball, the only possibly nonzero contributions to its recovery come from the opposite parity, so the assertion follows from the recovery equation for x. At a vertex v outside the ball, every neighbor u inside it must have distance exactly R from c:

$$R + 1 \le \operatorname{dist}(c, v) \le \operatorname{dist}(c, u) + 1 \le R + 1.$$

All these neighbors therefore have the same parity. One of the two sums is zero term by term; the other is the entire original recovery sum, which is zero because $x_v = 0$. This checks every boundary row, including its cancellations.

It follows that $Hs = s + As = s + t = x$. Every local codeword, and hence its span $\mathcal{L}_R(\mathcal{C})$, lies in $\operatorname{im} H$. Apply Lemma 9.1 to obtain (9.1). The quotient inequality is the second assertion of the same lemma. The split vectors need not themselves belong to the code, and we have not asserted $\operatorname{im} H \subseteq \mathcal{C}$. □

The dimension, quotient, and split-generator interfaces are formalized. The application to attained coordinate spaces, graph parity, and boundary cancellation above remains a written proof. At $R = 0$, a one-vertex-supported recoverable word is zero, so $\mathcal{L}_0(\mathcal{C}) = 0$, as the proof also shows. Zero-dimensional attained spaces and disconnected graphs cause no exception. Disjoint edges with equal values at their endpoints attain the half bound with radius-one generators. Conversely the triangle's parity code has dimension two on three scalar vertices and is generated inside radius-one balls; it violates half rate when the odd-cycle exclusion is removed.

Corollary 9.3 (Above-half codes need nonlocal dimension)**.** Not formalized; a complete proof is provided.

Put $D = \sum_v \dim_K U_v$. Under the hypotheses of Theorem 9.2, if $\dim_K \mathcal{C} \geq (1/2 + \delta)D$, then $\dim_K(\mathcal{C}/\mathcal{L}_R(\mathcal{C})) \geq 2\delta D$. In particular, for scalar characteristic-two codes on triangle-free graphs with N vertices and rate $1 - o(1)$, $\dim_K \mathcal{L}_1(\mathcal{C}) = o(N)$.

Proof (not formalized). Substitute the assumed dimension in the quotient inequality. For the scalar conclusion use $D \leq N$ and $\dim \mathcal{L}_1(\mathcal{C}) \leq D - \dim \mathcal{C} \leq N - \dim \mathcal{C} = o(N)$. □

Thus the known near-unit-rate triangle-free binary families [1, 11] necessarily have a negligible span of radius-one codewords. This consequence does not contradict local recovery: many global codewords can satisfy local equations without being sums of local codewords. For a construction, the theorem identifies a required nonlocal part of the code. For an impossibility argument, it leaves the obligation to prove local generation, together with the appropriate odd-cycle condition, for the actual code under study. Bounded causal depth alone supplies neither assertion.

9.1 Nonlinear sources and a recoverable affine hull

Local generation can be supplied by local changes to a nonlinear encoder. The additional requirement is recovery on the encoder's entire affine hull. Recovery on its image alone does not suffice.

Theorem 9.4 (Payment for exceptional sources)**.** Not formalized; a complete proof is provided.

Retain the graph, parent sets, and radius of Theorem 9.2, and suppose K is finite, with $q = |K|$. Let I be a finite set, let each A_i be finite and nonempty, and let

$$F : \prod_{i \in I} A_i \longrightarrow \bigoplus_v W_v, \qquad S = \operatorname{im} F, \qquad \operatorname{aff}_K S = a + \mathcal{C}.$$

Assume that each specified parent set recovers its coordinate on all of $a + \mathcal{C}$. Fix a set $B \subseteq I$ of exceptional sources. For each $j \notin B$, assume there is a center c_j such that changing only source j changes no output coordinate outside $B_R(c_j)$. Put

$$k = \dim_K \mathcal{C}, \quad r = \sum_v \dim_K \pi_v(\mathcal{C}), \quad \mathcal{L} = \mathcal{L}_R(\mathcal{C}), \quad \ell = \dim_K \mathcal{L}, \quad M = \prod_{i \in B} |A_i|.$$

Each fixed exceptional-source fibre has image in one affine coset of $\mathcal{L}$. Consequently,

$$2 \log_q |S| \leq r + \log_q M. \tag{9.2}$$

For any probability law of a source tuple Z on the stated Cartesian domain, with $X = F(Z)$ and entropies in bits,

$$2H(X) \leq r \log_2 q + H(Z_B). \tag{9.3}$$

Neither independence nor full support of this law is required. If $B = \varnothing$, then $\mathcal{C} = \mathcal{L}$ and $2k \le r$. If $S = a + \mathcal{C}$, then also

$$q^{\dim_K(\mathcal{C}/\mathcal{L})} \le M;$$

if, in addition, X is uniform on this affine space, then $\dim_K(\mathcal{C}/\mathcal{L}) \log_2 q \le H(Z_B)$.

Proof (not formalized). Translation by a transfers recovery from $a + \mathcal{C}$ to $\mathcal{C}$. Thus Theorem 9.2 gives $k + \ell \le r$. Two source assignments differing only in a nonexceptional coordinate have output difference in $\mathcal{C}$, supported in its specified radius-R ball, and hence in $\mathcal{L}$. Any two assignments with the same exceptional tuple can be joined by changing the remaining coordinates one at a time. All intermediate assignments are in the Cartesian domain. Summing the differences proves that their outputs lie in one affine $\mathcal{L}$-coset.

There are at most M such fibres, each with at most q^ℓ outputs, whereas $S \subseteq a + \mathcal{C}$ gives $|S| \le q^k$. Adding the logarithms of these two bounds and using $k + \ell \le r$ proves (9.2). For any source law the same support bounds give

$$H(X) \le k \log_2 q, \qquad H(X \mid Z_B) \le \ell \log_2 q.$$

Combine the second with $H(X) \le H(Z_B) + H(X \mid Z_B)$, add the first, and apply $k + \ell \le r$ to obtain (9.3).

If B is empty, choose $a \in S$. The coset assertion puts $S - a$ in $\mathcal{L}$. Since $S - a$ spans $\mathcal{C}$, this proves $\mathcal{C} = \mathcal{L}$. Finally, the quotient of $F(Z) - a$ modulo $\mathcal{L}$ is a function of Z_B. If $S = a + \mathcal{C}$, that function maps the exceptional-source domain onto $\mathcal{C}/\mathcal{L}$, proving the quotient cardinality bound. A uniform X induces the uniform law on this quotient, whose entropy is $(k - \ell) \log_2 q$. A function of Z_B has entropy at most $H(Z_B)$, proving the last assertion. □

The quantity r measures affine coordinate dimensions. The theorem does not replace it by $\sum_v H(X_v)/\log_2 q$ or by $\sum_v \log_q |\pi_v S|$. For uniform X on the entire affine hull, these quantities do agree with r. Empty source sets, singleton alphabets, constant encoders, and zero-dimensional coordinates are included. At radius zero, $\mathcal{L} = 0$, so nonexceptional sources cannot affect the output.

The all-local conclusion recovers the weighted half bound for locally generated linear codes: choose finitely many ball-supported generators, use their coefficients as the sources, and let F be their sum. The more general statement permits nonlinear source dependence and charges the actual exceptional tuple. To apply it to a native fibre, one must supply the Cartesian representation, the local influence sets, and recovery on the whole affine hull. A small circuit or a shallow recovery cone supplies none of these automatically.

Example 9.5 (Recovery can fail on the affine hull)**.** On the bipartite graph $K_{3,3}$, the binary encoder

$$(a, b, c) \longmapsto (a, b, c,\ c + ab,\ a + c + ab,\ b + c + ab)$$

has eight outputs and exact neighbor recovery. Writing the last three coordinates as y_1, y_2, y_3, the inverse formulas are $a = y_1 + y_2$, $b = y_1 + y_3$, and $c = y_1 + ab$. Each source affects only the radius-one star centered at its own left vertex. Nevertheless, the affine hull has dimension four, with basis

$$(1,0,0,0,1,0), \quad (0,1,0,0,0,1), \quad (0,0,1,1,1,1), \quad (0,0,0,1,1,1).$$

The first three vectors are outputs at the three unit sources; the fourth is $F(1,1,0) + F(1,0,0) + F(0,1,0)$. They span every output and are independent, as their first three coordinates show. Their span contains $(0,0,1,0,0,0)$, so the third coordinate is not recoverable there from its neighbors. All six coordinate dimensions are one; the inequality $2 \dim \mathcal{C} \le 6$ would fail if recovery on the affine hull were omitted. The original eight-word code still has exactly half rate.

10 Construction requirements and the next questions

The two groups of results constrain different objects. The cover theorems concern the coefficient algebra of a homogeneous check. The local theorem concerns the span of localized words in the code itself. The theorems do not supply either premise from the other. Their common use is to separate a promising construction from a family whose capacity is already controlled.

Hypothesis required of a candidate	Consequence here	What an above-half candidate must change
Fixed separated template, specified pro-p chain, homogeneous fixed widths	Half rate over every characteristic-p field	The representation, homogeneity, fixed-width condition, or code class.
Same template and chain, growing widths with tight simple-factor mass	Half rate; excess δ forces at least 2δ large-factor mass	A positive fraction of dimension in arbitrarily large absolute simple factors.
Independent-chord check, nonzero scalar edges, or invertible common-flag blocks	Rate tends to zero	Edge nonvanishing, the common flag, or the specified check family.
Characteristic two, odd girth greater than $2R+1$, radius-R local generation	Weighted half rate	A quantitatively nonlocal code subspace, or the graph premise.
Same graph, Cartesian sources with recoverable affine hull	$2H(X) \le r \log_2 q + H(Z_B)$	Sufficient exceptional-source entropy, or one of the representation premises.

Table 1: These are conditional tests, not automatic properties of tape computations.

10.1 A focused constructive target

The first target is a fixed separated voltage template and an actual chain of the stated type, with homogeneous coefficients whose geometric simple-factor dimensions grow. An above-half sequence would have to meet the mass lower bound of Corollary 5.2. Large factors alone are not evidence of high rate: the exact inflation control in Proposition 8.2 has them and still has a separate all-code half bound. A candidate should therefore be tested both for large factors and for an independent coding obstruction.

If the search instead uses characteristic-two codes on graphs of large odd girth, Corollary 9.3 gives a second test. Local recovery equations must leave enough independent global codewords outside the local span. A short description of the equations or a sparse check matrix is insufficient; the needed object is the actual kernel or subcode, with its dimension proved. The nonlinear-source extension offers a more concrete test: changing a local source must produce an actual local word of a recoverable affine hull. Recovery only on the encoder image cannot replace hull recovery. For group-word constructions on the cover families, increasing the alphabet size alone does not evade the bounds: homogeneous scalar exponents still give the uniform consequences of Corollary 7.2.

10.2 The corresponding impossibility targets

Question 10.1 (Unrestricted block width)**.** *Under the fixed template, separation, embedding, and chain hypotheses of Theorem 4.3, does half rate hold for every sequence of homogeneous characteristic-p checks, without a condition on their simple-factor dimensions?*

An affirmative answer would, by Proposition 8.1, also control arbitrary sheet-dependent linear checks on those same quotient relations. Conversely, such a sheet-dependent above-half check would inflate to a homogeneous counterexample with growing width. This is an equivalence within the specified cover model; it does not extend the template or group hypotheses.

Question 10.2 (A native source of the needed structure)**.** *Can actual shallow-cone fibres of a putative linear-time transposer be placed, with a fully charged cost, in one of the code classes controlled here or in an all-alphabet extension of those classes?*

The native paired-distance theorem of [5] alone does not give a fixed voltage template, homogeneous recovery, linearity of the fibre, an internal flag, or locally supported generators. These are the main missing implications. Even a successful coding step must be combined with a compatible cheap depth reduction and, for adaptive machines, a paid description of the input-dependent trace or relation.

A Index of Lean declarations

Declarations are relative to `CoverStorage`; their axiom checks are generated together with this index (Appendix B).

`CoverStorage.HomogeneousInflation` `finrank_kernel_inflated`, `inflatedCheck`, `kernelEquiv`, `rate_cross_product`, `reindex_inflated`, `sheetCheck`, `sheetReindex`.

`CoverStorage.LocalGeneration` `kernel_range_dimension_bound`, `quotient_dimension_bound`, `split_generator_dimension_bound`.

B Proof status and verification

The Lean-verified results are exact homogenization (Proposition 8.1) and the dimension interfaces (Lemma 9.1). The shared `LeanMachine/HomogeneousInflation.lean` module constructs the two checks and their coordinate permutation, proves the kernel equivalence, and applies Mathlib's finite-product dimension formula. `LeanMachine/KernelRangeBound.lean` reuses Mathlib's dimension monotonicity, rank-nullity, quotient dimension, and span closure for the dimension, quotient-deficit, and split-generator bounds. The paper modules expose these results without duplicating their proofs. The English outlines are guides; the Lean declarations are authoritative.

The remaining graph, asymptotic cover, nonlinear-source, and group-word results have complete written proofs. In particular, the local-generation theorem is not described as formalized merely because its final dimension inference is checked. The native hypotheses, free-group separation, imported Cohn and Jaikin-Zapirain–Souza theorems, parameter uniformization, coefficient filtrations, weighted determinant argument, and full-matrix-algebra inflation example are not formalized in this manuscript. Nor is the written product-subcode corollary part of the formal kernel equivalence. The source-coset argument and central-layer group-word specialization are also written proofs, not additional claims about the scope of that interface. The README's formalization todos record the remaining dependencies and existing library material to reuse. No numerical computation is used as a proof premise.

Lean and Mathlib are pinned to v4.34.1. The paper requires the sibling shared package through a relative path. Clone the whole repository with its directories intact. From the paper directory, `python3 tools/lean_index.py` generates the declaration index and `lean/PaperAxioms.lean`

from the same annotations. In `lean/`, run `lake exe cache get`, `lake build`, and `lake env lean PaperAxioms.lean`. The cited declarations use at most `propext`, `Classical.choice`, and `Quot.sound`. In `paper/`, run `latexmk -pdf main.tex`.

References

[1] Alexander Barg, Moshe Schwartz, and Lev Yohananov. Storage codes on coset graphs with asymptotically unit rate. arXiv:2212.12117v3, 2024. Version dated 12 June 2024. URL: https://arxiv.org/abs/2212.12117v3.

[2] Alexander Barg and Gilles Zémor. High-rate storage codes on triangle-free graphs. *IEEE Transactions on Information Theory*, 68(12):7787–7797, 2022. URL: https://arxiv.org/abs/2110.02378v2, doi:10.1109/TIT.2022.3191309.

[3] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and partial Lean formalization, repository HansBrende/papers, 2026. Pinned revision e5e03d2cdbc566d5816b682f7d2549a694b8cad9. URL: https://github.com/HansBrende/papers/tree/e5e03d2cdbc566d5816b682f7d2549a694b8cad9/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[4] Hans Brende and Claude Opus 5.5 (Anthropic). Matrix transposition and the tape coding conjecture. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision e5e03d2cdbc566d5816b682f7d2549a694b8cad9. URL: https://github.com/HansBrende/papers/tree/e5e03d2cdbc566d5816b682f7d2549a694b8cad9/Matrix-Transposition-and-the-Tape-Coding-Conjecture.

[5] Hans Brende and GPT-6 Astra (OpenAI). Directed recovery structures in tape computations. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision e5e03d2cdbc566d5816b682f7d2549a694b8cad9. URL: https://github.com/HansBrende/papers/tree/e5e03d2cdbc566d5816b682f7d2549a694b8cad9/Directed-Recovery-Structures-in-Tape-Computations.

[6] Hans Brende and GPT-6 Astra (OpenAI). Local coding barriers and the limits of tape geometry for matrix transposition. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision e5e03d2cdbc566d5816b682f7d2549a694b8cad9. URL: https://github.com/HansBrende/papers/tree/e5e03d2cdbc566d5816b682f7d2549a694b8cad9/Local-Coding-Barriers-and-the-Limits-of-Tape-Geometry-for-Matrix-Transposition.

[7] P. M. Cohn. *Free Ideal Rings and Localization in General Rings*, volume 3 of *New Mathematical Monographs*. Cambridge University Press, 2006. The free-group Sylvester-domain result is also stated in the introduction of Jaikin-Zapirain and Souza. doi:10.1017/CBO9780511542794.

[8] Andrei Jaikin-Zapirain and Henrique Souza. Sylvester domains and pro-p groups. *Documenta Mathematica*, 31(2):453–502, 2026. URL: https://ems.press/journals/dm/articles/14299124, doi:10.4171/DM/1034.

[9] John R. Stallings. Topology of finite graphs. *Inventiones Mathematicae*, 71(3):551–565, 1983. doi:10.1007/BF02095993.

[10] The Stacks Project Authors. Hilbert nullstellensatz. The Stacks Project, Theorem 10.34.1, tag 00FV, 2026. Accessed 25 October 2026. URL: https://stacks.math.columbia.edu/tag/00FV.

[11] Aryeh Lev Zabokritskiy. Asymptotically unit-rate storage codes from binary BCH codes. arXiv:2610.01733v1, 2026. Author also identified as Yohananov; version dated 1 October 2026. URL: https://arxiv.org/abs/2610.01733v1.